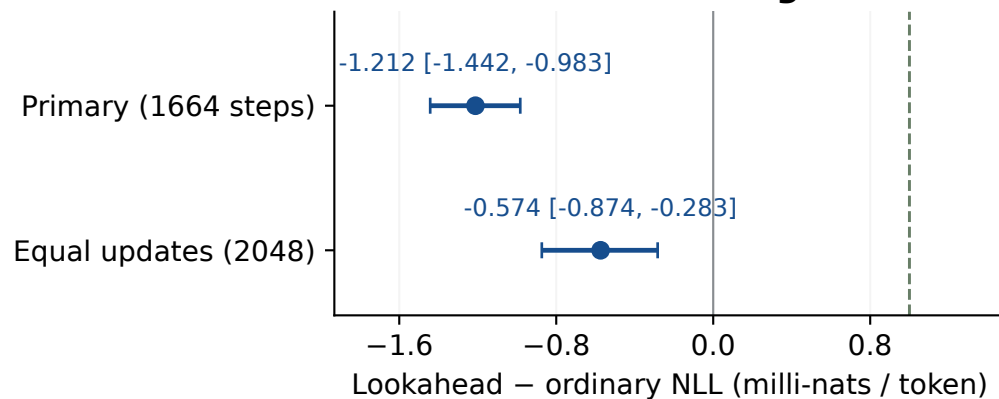


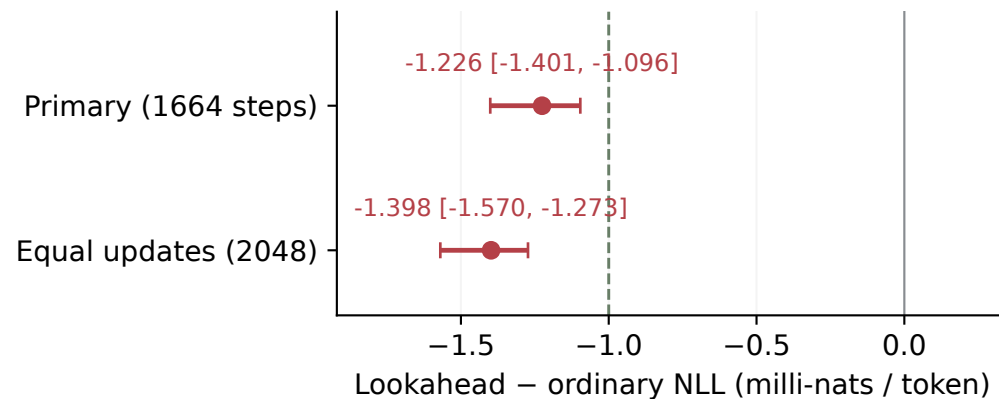
Fixed Lookahead comparison on unused documents

512 new documents per domain · candidate and controls fixed before scoring · seed 0

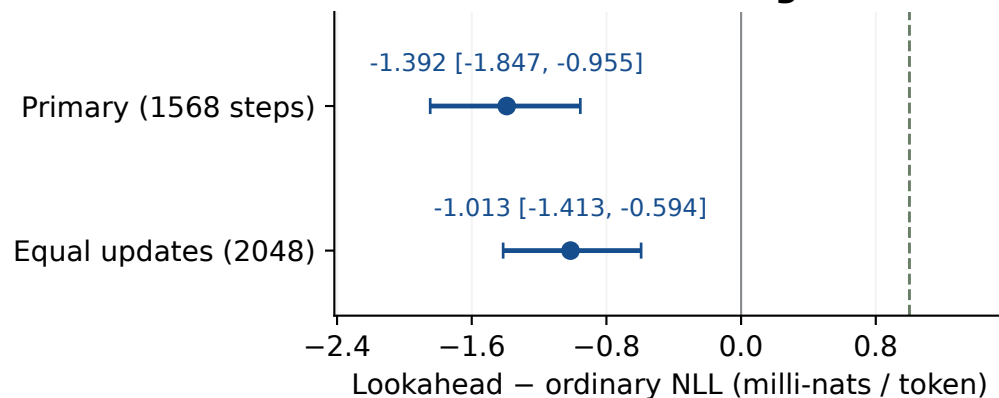
A 135M · Math learning



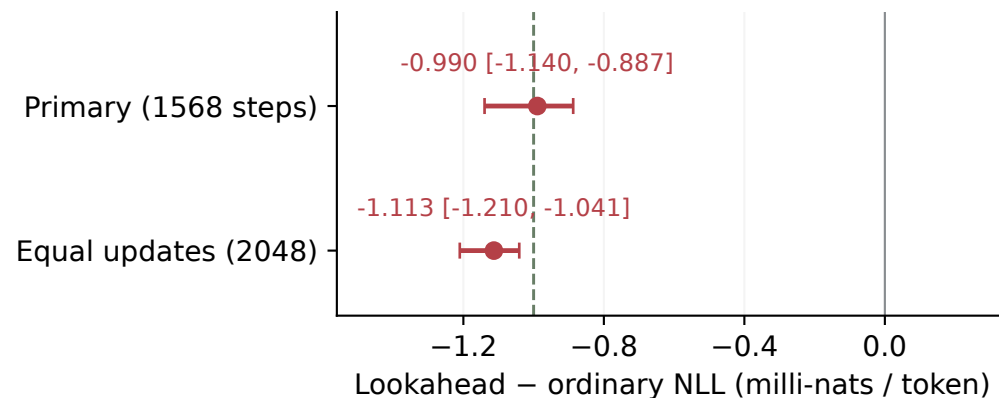
B 135M · FineWeb retention



C 360M · Math learning



D 360M · FineWeb retention



Lower is better. Bars: paired-document 95% intervals; they do not measure training-seed variation.

Dashed lines: +0.001 math margin / -0.001 retention effect threshold. Primary candidate uses more updates.

Known Lookahead-style interpolation; the normalized T³ wave-2a null is unchanged.